
Sprint 1 Review

SE3022 · Case Study Project · BSc (Hons) Computer Science
Year 3, Semester 1 · 2026

GROUP 06

Sprint dates · 19–30 August 2026

Project Overview

A1 Academy — a centralised Learning Management System for tuition centres.

THE PROBLEM

Traditional tuition management is fragmented — manual attendance, scattered materials, and no clear line of sight into student progress for teachers or admins.

OUR SOLUTION

A unified, secure LMS connecting Administrators, Teachers and Students — each with a role-based interface — around one centralised hub for materials, assignments and announcements.

WHO IT SERVES

Teachers

Class scheduling, digital materials, attendance and engagement tools.

Students

Course catalogue, enrolment, submissions and visible progress.

Administrators

Platform oversight, quality control and security governance.

Agenda

01

Project Overview

What we're building, and for whom

03

Individual Contributions

What each of us delivered, in detail

05

Backlog & Planning

13 issues, groomed and estimated in JIRA

07

Feature Delivery – Live Demo

Authentication & role-based access control

09

Sprint Metrics

Velocity, completion rate, workload split

11

Committed vs. Delivered

Planning promises, checked against the review

13

Sprint 2 Kickoff

Already underway since 3 September

02

Team & Roles

Who owned what this sprint

04

Sprint Goal & Scope

What we committed to, and why

06

Platform Foundation

Architecture, database, event streaming, CI/CD

08

Quality Engineering

Unit, integration, UI, E2E & load testing

10

Sprint Burndown

Story points remaining, tracked daily

12

Retrospective

What we'll change going into Sprint 2

14

Questions

Open floor

Team & Roles

Sprint 1 responsibilities across development, infrastructure and quality assurance.



DEVELOPER

Heshanth

IT24101633

AA-12 · AA-13 · AA-14 · AA-15 · AA-16



BUSINESS ANALYST

Akash

IT24101853

AA-9



DEVOPS ENGINEER

Piranavan

IT24101506

AA-10 · AA-11



QA ENGINEER

Inkaran

IT24102182

AA-17 · AA-18 · AA-19 · AA-20 · AA-21

🧑‍🤝‍🧑 Though split by primary role, all four collaborated across code review, planning and testing throughout the sprint.

Sprint 1 Contribution – Heshanth T

SECURE AUTHENTICATION & ROLE-BASED ACCESS IMPLEMENTATION

- Built the student and teacher registration endpoints with live email-uniqueness checks.
- Developed the login endpoint, issuing signed JWTs carrying role claims.
- Implemented role-based access control across navigation and protected API routes.
- Engineered secure password hashing for every stored credential.
- Built the self-service password reset flow with a time-limited reset link.
- Worked with QA to resolve issues surfaced during integration testing.
- Prepared and rehearsed the live feature demo for this review.

ASSIGNED TEAM ROLE

Developer

IT24101633

SPRINT 1 DIRECT IMPACT

Delivered the secure authentication gateway and access-control logic that every other Sprint 1 feature depends on.

Sprint 1 Contribution — Akash D

REQUIREMENTS ENGINEERING & BACKLOG GROOMING

- Analysed the case study brief and translated it into Sprint 1 requirements.
- Wrote and refined user stories for AA-12 through AA-16 in the Jira backlog.
- Defined Given/When/Then acceptance criteria for every Sprint 1 story.
- Organised and maintained the 13-item backlog across both Sprint 1 epics.
- Supported story-point estimation — all 13 estimates logged within five minutes of sprint start.
- Scaffolded the initial repository and branching strategy (AA-9).
- Helped prepare this Sprint 1 Review and groom the early Sprint 2 backlog.

ASSIGNED TEAM ROLE

Business Analyst

IT24101853

SPRINT 1 DIRECT IMPACT

Ensured the team worked from clear, testable requirements and a well-groomed backlog throughout the sprint.

Sprint 1 Contribution – Piranavan A

CI/CD PIPELINE AUTOMATION & INFRASTRUCTURE SETUP

- Provisioned the PostgreSQL database and Apache Kafka event broker.
- Built the GitHub Actions pipeline for automated build, test, and containerisation.
- Configured branch protections requiring passing checks before merge.
- Set up automated deployment to Azure App Service.
- Kept every Sprint 1 feature branch at 3/3 passing checks before merge into main.
- Diagnosed and resolved pipeline and deployment configuration issues.
- Supported the team's day-to-day environment setup and developer workflow.

ASSIGNED TEAM ROLE

DevOps Engineer

IT24101506

SPRINT 1 DIRECT IMPACT

Established the pipeline infrastructure that let every Sprint 1 change be built, tested, and deployed automatically.

Sprint 1 Contribution – Inkaran K

QUALITY ENGINEERING ACROSS FIVE TEST LAYERS

- Designed unit tests covering the authentication logic (AA-17).
- Built integration tests exercising every /api/auth endpoint through the real pipeline (AA-18).
- Wrote frontend component tests for registration and login (AA-19).
- Automated end-to-end browser testing of the full auth flow with Selenium (AA-20).
- Load-tested the login endpoint up to 200 concurrent users (AA-21).
- Logged defects and coverage gaps for Sprint 2 follow-up.
- Verified the account approval, rejection, and deactivation state transitions.

ASSIGNED TEAM ROLE

QA Engineer

IT24102182

SPRINT 1 DIRECT IMPACT

Verified that every Sprint 1 story met its acceptance criteria and passed automated checks at every layer before this review.

Sprint Goal & Scope

SPRINT GOAL

Stand up the platform's technical foundation and deliver secure, role-based authentication for Students, Teachers, and Administrators.



AA-1 · Initial Setup, Architecture & Deployment
Repository, database, event streaming, CI/CD pipeline

3 tasks



AA-2 · Identity & Authentication Service
Registration, login, RBAC, password reset — plus full test coverage

10 items

SPRINT AT A GLANCE

2

Epics in progress

13

Issues committed

30

Story points committed

19–30 Aug

Sprint window (2 weeks)

Sprint Backlog & Planning

13 issues

30 story points

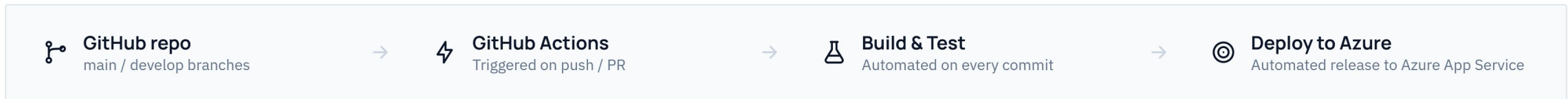
100% groomed & estimated

KEY	SUMMARY	TYPE	SP	ASSIGNEE	STATUS
AA-1 · INITIAL SETUP, ARCHITECTURE & DEPLOYMENT					
AA-9	Repository & Application Initialization	Task	1	Akash	Done
AA-10	Database & Event Broker Provisioning	Task	3	Pirnavan	Done
AA-11	CI/CD Pipeline & Deployment Automation	Task	5	Pirnavan	Done
AA-2 · IDENTITY & AUTHENTICATION SERVICE					
AA-12	Student Registration	Story	3	Heshanth	Done
AA-13	Teacher Registration	Story	2	Heshanth	Done
AA-14	User Login	Story	3	Heshanth	Done
AA-15	Role Based Access Control	Story	2	Heshanth	Done
AA-16	Password Reset	Story	2	Heshanth	Done
AA-17	Unit Testing — Authentication Logic	Task	1	Inkaran	Done
AA-18	API Integration Testing — Role Enforcement	Task	2	Inkaran	Done
AA-19	Frontend Component Validation Testing	Task	2	Inkaran	Done
AA-20	End-to-End Authentication Flow Verification	Task	3	Inkaran	Done
AA-21	Load Testing — Login Endpoint Performance	Task	1	Inkaran	Done

Platform Foundation



CONTINUOUS INTEGRATION & DELIVERY



EVERY BRANCH, CHECKED BEFORE MERGE

The screenshot shows the GitHub Branches page with the following data:

Branch	Updated	Check status	Behind Ahead	Pull request
main	2 days ago	✓ 5 / 5	Default	
feature/User-Deactivation	3 days ago	✓ 3 / 3	1 0	#16
feature/Teacher-Approval	3 days ago	✓ 3 / 3	3 0	#15
feature/Filter-Pending-Teachers	3 days ago	✓ 3 / 3	5 0	#14

Every feature branch this sprint carried its own 3/3 checks — build, test, and containerize — before merging into **main**, which sits at 5/5 with zero branches behind on unmerged work.

Feature Delivery – Auth & RBAC

[LIVE DEMO](#)

AA-12

Student Registration

Signup with live email-uniqueness checks and bcrypt password hashing.



AA-13

Teacher Registration

Qualifications submitted; account held Pending until an admin approves it.



AA-14

User Login

JWT issued on verified credentials — powers every protected session.



AA-15

Role Based Access Control

Nav hidden by role client-side; API still returns 403 on unauthorized calls.



AA-16

Password Reset

Self-service recovery through a securely issued, time-limited reset link.

WHAT WE'LL SHOW LIVE

1

Sign up



2

Teacher held Pending



3

Log in · JWT issued



4

Role-based dashboard



5

Forgot & reset password

Quality Engineering

Five layers of automated coverage, from a single function up to the whole login endpoint under load.

1	 Unit Testing AA-17	Password hashing verified correct & salted; JWT payload claims (role, subject, expiry) validated.	Code-level, isolated
2	 Integration Testing AA-18	Every /api/auth endpoint exercised through the real request pipeline — not mocked.	200 · 400 · 401 · 403
3	 Frontend Validation AA-19	React form validation on signup & login; JWT stored and read safely from localStorage.	3 components covered
4	 E2E · Selenium AA-20	Full browser automation, no mocks: registration through to dashboard access.	Signup-to-dashboard
5	 Load Testing · JMeter AA-21	Concurrent login requests simulated against the endpoint under stress.	< 5s under load

✓ All five layers passing before this review — unit through load, with no manual-only checks standing in for automation.

Sprint Metrics

13 / 13

Issues completed

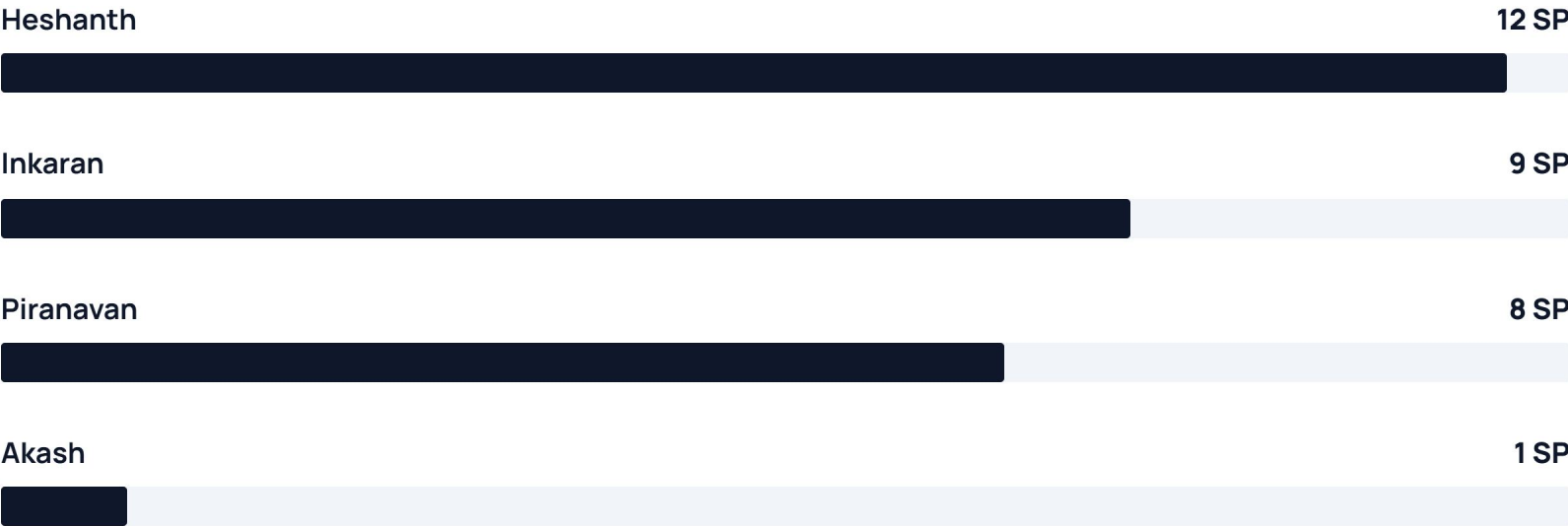
30 / 30

Story points delivered

100%

Completion rate · zero carryover

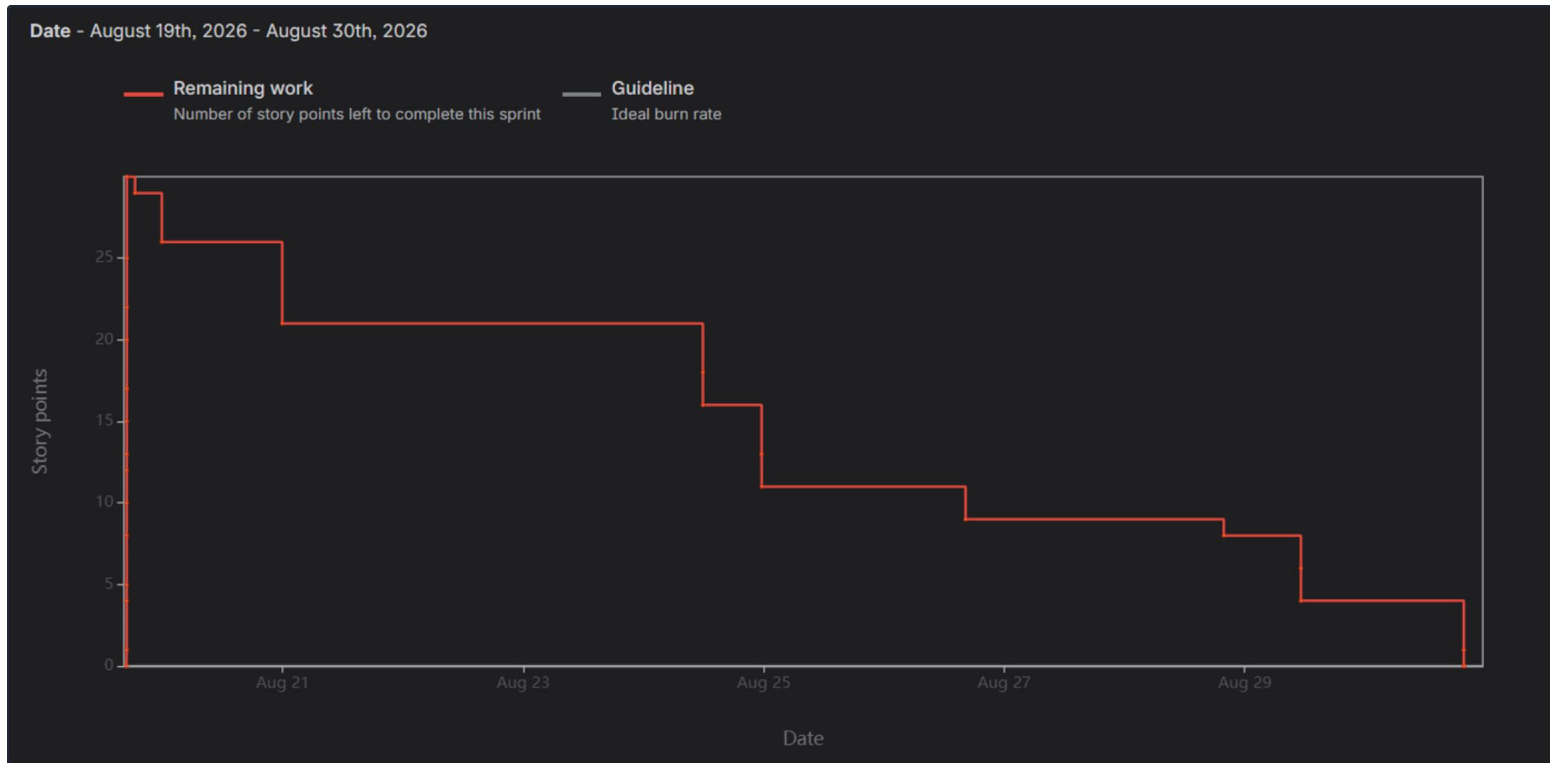
STORY POINTS DELIVERED BY TEAMMATE



⌚ Distribution reflects role emphasis this sprint — Heshanth carried the feature build, Inkaran the full QA suite. Revisited in the retrospective.

Sprint Burndown

Story points remaining, tracked daily against the sprint window.



Source: Jira · AA Sprint 1 board · burndown report

BURNDOWN AT A GLANCE

30 → 0

Story points burned down over the sprint

Infrastructure first

AA-9–AA-11 — repo, database, and CI/CD — closed within the first day and a half.

Zero carryover

Final ticket closed 7:18pm on Aug 30 — all 30 committed points delivered on time.

Committed vs. Delivered

Every item promised at Sprint Planning, checked against what was actually shown at this review.

🕒 Committed at Planning

Stand up the core architecture — React frontend, ASP.NET Web API backend, PostgreSQL, Kafka.

Configure a GitHub Actions pipeline for automated build and test on every push.

Deliver secure registration, login and role-based access for Students, Teachers and Admins.

Back every story with full automated test coverage before the sprint closes.

✅ Demonstrated at Review

Live end-to-end flow: sign up → teacher held Pending → log in → role-based dashboard.

GitHub Actions running build and test automatically on every commit and PR.

Self-service password reset through a securely issued, time-limited link.

All five test layers passing — unit, integration, frontend, E2E and load.

✅ Every committed item shipped and demonstrated live — no descoped stories, no carryover into Sprint 2.

Retrospective

✔ What Went Well

Test coverage was built in from day one — unit through load — not bolted on afterward.

Infrastructure and feature work ran in parallel with no blocking dependencies between them.

100% of committed scope delivered — every one of 13 issues, zero carryover.

⚠ What Could Improve

Story points landed unevenly across the team — 12 for one person, 1 for another.

Testing (AA-17–21) didn't start until day 9 of an 11-day sprint, once development was feature-complete.

No mid-sprint check-in existed to catch scope or estimation drift early.

🔄 Action Items · Sprint 2

Rebalance ticket sizing at planning so estimates map more evenly across the team.

OWNER: WHOLE TEAM, AT PLANNING

Bring QA into each story from day one instead of testing after development finishes.

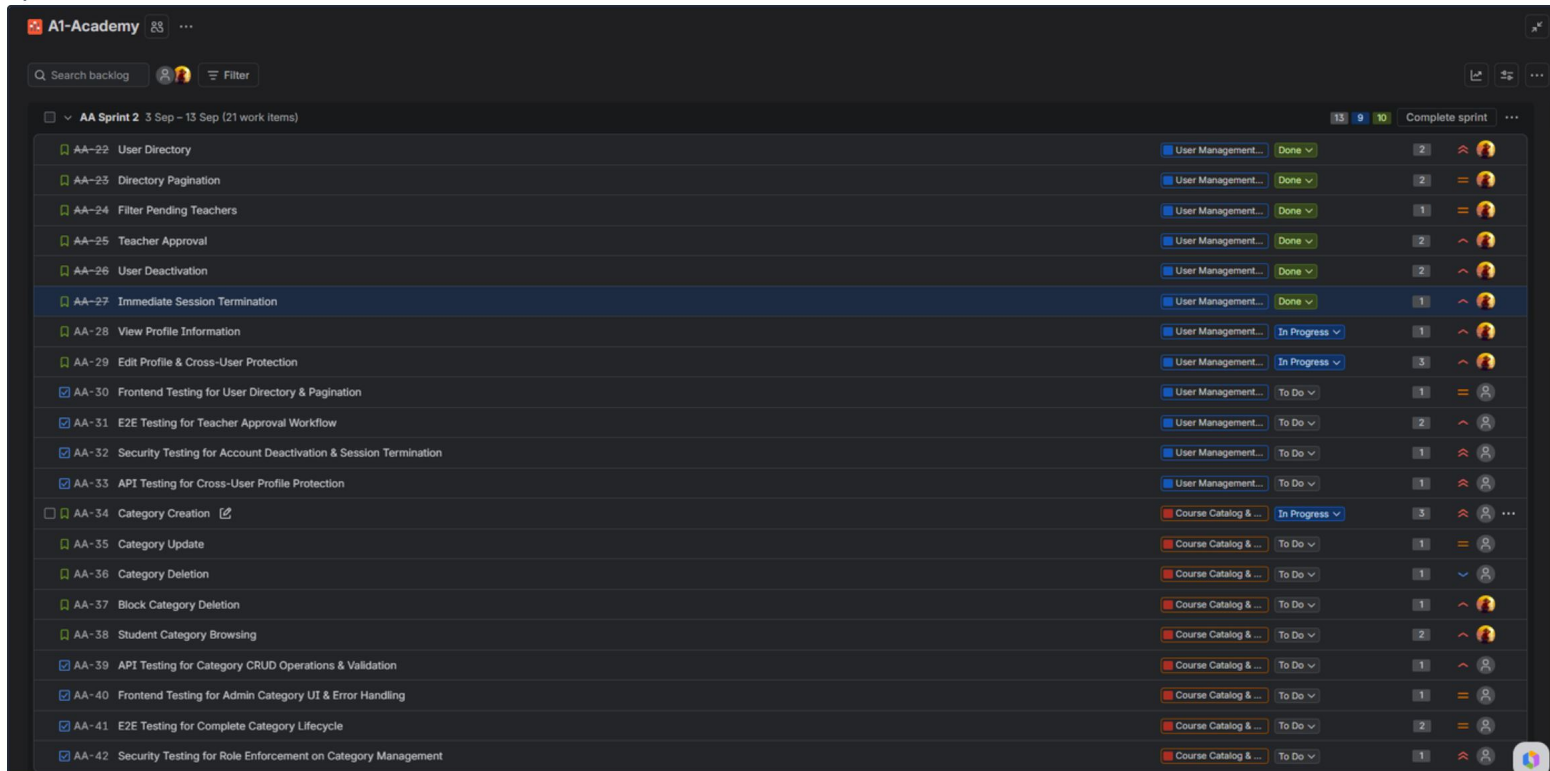
OWNER: INKARAN & HESHANTH

Add a short mid-sprint check-in to catch drift while there's still time to react.

OWNER: PIRANAVAN

Sprint 2 – Already Underway

Started the day after Sprint 1 closed — no gap between review and the next sprint.



Source: Jira · AA Sprint 2 board · 3–13 September 2026

SPRINT 2 AT A GLANCE

21 items

Across 2 epics — User Management & Moderation, Course Catalog & Category Service

6 **3** **12**
Done In progress To do

Already shipped this sprint

Teacher Approval and User Deactivation are both Done — live, tested, and deployed via the same CI/CD pipeline as Sprint 1.

Thank You

Questions & discussion

Heshanth
Developer

Akash
Business Analyst

Pirnavan
DevOps

Inkaran
QA